Pension Receipt and Depression among Older Adults in India*

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Abstract

This paper seeks to answer whether older adults who receive a pension are less likely to be depressed. Using Wave 1 of the Longitudinal Ageing Study in India (2017–18), I study 29,627 adults aged 60 and above, a population in which 9.2 percent meet the CIDI-SF criteria for probable major depression. Conditional on demographic and socioeconomic controls and state fixed effects, the estimated difference in depression between pension recipients and non-recipients is small and insignificant (0.55 percentage points; 95 percent confidence interval −0.13 to 1.24), which rules out any meaningful protective association. There's a sharp difference between the recipients of the two types of pensions: work-related pensions are received by more educated, urban and better-off adults, while social pensions go to poor households and widows. The significant positive associations are concentrated in the national old-age pension, whose eligibility condition, a below-poverty-line ration card, itself is a predictor of depression. Cross-sectional comparisons of pension recipients therefore cannot show whether pensions improve mental health.

Keywords: pensions, depression, ageing, social protection, India
JEL codes: H55, I12, I38, J14

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1 Introduction

As life expectancy in India rises and the share of older adults grows, income security, health, and well-being in later life have become central questions for public policy. Pensions are the main instrument designed to support older citizens: they provide a stable income after work ends and reduce vulnerability in old age. Beyond this financial role, pension income may also affect mental health, by lowering financial stress and reducing dependence on family members. In LASI, 34.7 percent of adults aged 60 and above receive a pension, and about five in six of them receive a social pension rather than a work-related one (Table 1).

Depression among older adults is common but it receives little policy attention. In later life it is linked to declining physical health, social isolation, widowhood, reduced mobility, and economic insecurity. In LASI, 9.2 percent of adults aged 60 and above meet the criteria for probable major depression. For many older Indians, especially in poorer households, mental well-being depends not only on health but on household structure, social status, and whether they have a regular income, which makes pension receipt a natural candidate for study.

The fundamental question this paper asks is whether older adults who receive a pension are less likely to be depressed. The question is hard to answer from survey data, because pension receipt is not random and the two main kinds of pension in India reach very different people. Work-related pensions go to former formal-sector workers, who are more educated, more urban and better off than other older adults. Social pensions are targeted at the disadvantaged: the National Old Age Pension Scheme requires a below-poverty-line household and the widow pension requires widowhood, and both conditions are predictors of depression. Selection therefore runs in opposite directions for the two kinds of pension. Comparing work-pension recipients with non-recipients is likely to overstate any benefit of pensions, while comparing social-pension recipients with non-recipients is likely to understate it, and a comparison that combines the two mixes both biases.

I study this question using Wave 1 of the Longitudinal Ageing Study in India (LASI), which records pension receipt and depressive symptoms for 29,627 adults aged 60 and above, and I measure probable major depression with the CIDI-SF scale. Conditional on demographic and socioeconomic controls and state fixed effects, the estimated difference in depression between pension recipients and otherwise similar non-recipients is 0.55 percentage points and not statistically significant (95 percent confidence interval -0.13 to 1.24). The interval rules out a protective association larger than 0.13 percentage points, about 2 percent of the sample mean of 7.6 percent. Separating the two kinds of pension, social-pension recipients are 0.70 percentage points more likely to be depressed (significant at the 10 percent level), and among social pensions the national old-age pension shows the largest estimate, 0.82 percentage points (significant at the 5 percent level). The estimate for work-related pensions is too imprecise to be informative. Holding a BPL ration card, the eligibility condition for the old-age pension, is itself associated with a 1.57 percentage point higher probability of depression, and with that controlled, the social-pension estimate falls to 0.57 and is no longer significant. The conclusion that the data rule out a meaningful protective association holds across alternative standard errors, pension definitions, a probit model and survey weights.

The paper relates to work on old-age pensions and the welfare of the elderly and their households. Case and Deaton (1998) study South Africa's large old-age pension and how the income is spent within households, while Kaushal (2014) and Unnikrishnan and Imai (2020) study India's old-age pension and its effects on the labour supply and well-being of the elderly and on household welfare. The paper that is closest in outcome is Galiani et al. (2016), who use age and geographic eligibility cutoffs in Mexico and find that a non-contributory pension lowers depression scores among older adults. This paper adds evidence from India's national ageing survey on how selection into pensions shapes naive comparisons: because work-related and social pensions select recipients in opposite directions, a cross-sectional comparison can show no protective association even if pensions do improve mental health, as the Mexican evidence suggests. Section 2 describes the data and Section 3 the empirical approach. Section 4 presents the results, Section 5 discusses how to interpret them, and Section 6 concludes.

2 Data

2.1 The Longitudinal Ageing Study in India

The analysis uses Wave 1 of the Longitudinal Ageing Study in India (LASI), fielded in 2017–18 (IIPS et al., 2020b; IIPS, 2020a). LASI is a nationally representative survey of adults aged 45 and above and their spouses, covering every state and union territory except Sikkim. Its questionnaire is harmonised with the Health and Retirement Study and covers demographics, work and retirement, pensions and social welfare schemes, health, and mental health. Because it records depressive symptoms and pension receipt for the same respondents, it is well suited to this question.

2.2 Sample

The population of interest is adults aged 60 and above, the group to whom LASI's social-pension questions are addressed. Wave 1 interviewed 72,250 individuals, of whom 31,464 were aged 60 or above. I exclude 818 respondents who were not administered the depression module (both screening questions missing, most likely proxy interviews), 1,018 with missing caste, and one with missing religion. The estimation sample contains 29,627 respondents in 22,700 households.

2.3 Measuring depression

The outcome is an indicator of probable major depression constructed from the Composite International Diagnostic Interview – Short Form (CIDI-SF; Kessler et al., 1998), administered in questions MH201–MH223. The module has two screening pathways. The first begins with MH201, which asks whether the respondent felt sad, blue, or depressed for two weeks or more in a row during the past year; the second begins with MH214, which asks about a two-week period of lost interest in most things. In each pathway, the symptom questions are asked only if the episode lasted most of the day, nearly every day (MH202–MH203 and MH215–MH216). The depressed-mood pathway

records seven symptoms (MH204–MH211) and the loss-of-interest pathway six (MH217–MH222); Appendix B lists them.

A respondent is classified as depressed if either pathway records three or more symptoms:

$$\text{Depressed}_i = \mathbf{1}\big[\max(\text{MoodSymptoms}_i,\ \text{InterestSymptoms}_i) \geq 3\big].$$

Because the symptom questions are reached only through a positive screen that meets the duration criteria, this preserves the two-stage structure of the CIDI-SF and avoids classifying respondents as depressed on the basis of isolated symptoms such as fatigue or poor sleep. Respondents who were never administered the module are treated as missing rather than as not depressed. In the estimation sample, 7.6 percent of respondents are classified as depressed, or 9.2 percent using LASI's national weights.

2.4 Measuring pension receipt

LASI records pensions in two places. The work, retirement and pension module asks respondents who work in, or have retired from, the organized sector whether they currently receive a pension or expect one in future (WE412). The social welfare module asks every respondent aged 60 and above, scheme by scheme, whether they are aware of it (SW201) and whether they currently avail its benefits (SW202). The cash schemes are the National Old Age Pension Scheme, which requires a below-poverty-line household; the widow pension, asked only of women; and other state or central pension schemes. I exclude the Annapurna scheme, which provides food grains rather than cash.

The main treatment is current receipt of any pension:

$$\text{Pension}_i = \mathbf{1}\big[\text{WE412}_i = 1 \ \text{ or } \ \text{SW202}_{s,i} = 1 \text{ for some cash pension scheme } s\big].$$

Respondents who were not asked a question, because they never worked in the organized sector or were unaware of a scheme, are coded as not receiving that pension. A broader definition that adds respondents who expect a work-related pension in future (WE412 = 2) is used only as a robustness check, since they receive no pension income yet. Because the two kinds of pension reach very different people (Section 4.1), I also report work-related and social pensions separately, and social pensions by scheme.

2.5 Controls

The regressions control for characteristics that plausibly affect both pension receipt and mental health: age and its square; sex; household structure; urban residence; years of schooling (zero for respondents who never attended school); caste (Scheduled Caste, Scheduled Tribe, Other Backward Class, and none of these, with OBC as the reference group); religion (Hindu, Muslim, Christian, Sikh, and other, grouping categories with very few respondents); quintiles of monthly per capita consumption expenditure (MPCE); and state fixed effects.

Household structure combines marital status and living arrangements into seven categories: married and living with a spouse only, or with a spouse and children; married with the spouse not co-resident; widowed and living alone, with children, or with others; and other unmarried respondents. The two variables cannot usefully enter separately, because no widowed respondent lives with a spouse: with both in the regression, the contrast between widowed and married respondents would be identified only from the 212 married respondents who live apart from their spouse.

Finally, to capture the eligibility rule for the old-age pension directly, I merge from the household questionnaire whether the household holds a below-poverty-line (BPL) or Antyodaya ration card (CO301–CO302).

3 Empirical approach

I estimate linear probability models of the form

$$100 \times \text{Depressed}_i = \beta\, \text{Pension}_i + \mathbf{X}_i'\gamma + \mu_{s(i)} + \varepsilon_i, \tag{1}$$

where $\mathbf{X}_i$ contains the controls described above and $\mu_{s(i)}$ is a fixed effect for respondent i's state. Scaling the outcome by 100 makes β the difference in the probability of depression, in percentage points, between pension recipients and otherwise similar non-recipients. Table 2 adds the controls in four steps, from no controls to the preferred specification in column 5. State fixed effects absorb all state-level differences, including in pension coverage, benefit generosity, and health systems, so the preferred estimate compares recipients and non-recipients within the same state.

Three choices deserve comment. First, MPCE and household structure are measured after pension receipt and may partly respond to it: a pension raises household spending (Kaushal, 2014) and can change whom an older person lives with. Controlling for them may therefore absorb part of any effect (Angrist and Pischke, 2009), so Appendix Table 4 also reports estimates without them. Second, several respondents are interviewed in each household, so standard errors are clustered by household; clustering instead by the village or urban ward from which LASI sampled households gives almost identical standard errors (0.36 against 0.35 for the preferred estimate; Appendix Table 4, column 3). Because the outcome is binary, the errors are heteroskedastic by construction, and the clustered standard errors are robust to this. Third, the regressions are unweighted, since the controls include LASI's main design variables (state, urban residence, and age); estimates using LASI's national weights are reported as a robustness check (Solon et al., 2015). Descriptive statistics are weighted.

Table 1: Characteristics of respondents aged 60+ by pension status

	All	No pension	Work pension	Social pension
Depressed (CIDI-SF)	0.092	0.088	0.070	0.105
Receives any pension	0.347	0.000	1.000	1.000
Work-related pension	0.057	0.000	1.000	0.008
Social pension	0.292	0.000	0.041	1.000
Age (years)	68.993	68.221	69.922	70.572
Female	0.527	0.534	0.121	0.588
Widowed	0.359	0.312	0.188	0.499
Lives alone	0.057	0.043	0.041	0.092
Urban	0.288	0.309	0.548	0.188
Years of schooling	3.253	3.393	10.260	1.595
Scheduled Caste	0.195	0.176	0.144	0.248
Scheduled Tribe	0.083	0.076	0.051	0.104
BPL or Antyodaya card	0.535	0.504	0.213	0.667
Bottom two MPCE quintiles	0.435	0.428	0.253	0.487
Observations	29,627	18,118	2,057	9,539

Means weighted by LASI national individual weights; observations unweighted.

Work- and social-pension columns overlap for respondents receiving both.

4 Results

4.1 Who receives a pension?

Table 1 reports weighted means of respondents' characteristics by pension status. Of adults aged 60 and above, 34.7 percent receive a pension. Most of this is accounted for by social pensions: 29.2 percent receive a social pension and 5.7 percent a work pension (a handful receive both), so about five in six recipients receive a social pension. Work-pension recipients are better off than non-recipients on almost every measure. They have about three times as much schooling (10.3 years against 3.4), and 55 percent live in urban areas, compared with 31 percent of non-recipients. Only 21 percent hold a BPL card, against 50 percent of non-recipients, and only 25 percent are in the bottom two MPCE quintiles, against 43 percent. They are also less likely to be widowed (19 against 31 percent) and far less likely to be women (12 against 53 percent), most likely because the formal-sector jobs that carry work pensions were held mostly by men. For social-pension recipients the situation is reversed: they are worse off than non-recipients. The corresponding figures are 1.6 against 3.4 years of schooling, 19 against 31 percent for urban residence, 67 against 50 percent for holding a BPL card, and 49 against 43 percent for being in the bottom two MPCE quintiles. Half are widowed, against 31 percent of non-recipients, and 59 percent are women, against 53 percent. This reversal is what the eligibility rules imply: the old-age pension requires a BPL household and the widow pension requires widowhood. The two groups therefore differ from non-recipients in opposite directions on every one of these characteristics.

The raw depression rates are 8.8 percent among non-recipients, 7.0 percent among work-pension

recipients and 10.5 percent among social-pension recipients. The overall depression rate is 9.2 percent. The better-off group (work-pension recipients) is less depressed and the worse-off group (social-pension recipients) is more depressed. So clearly these raw gaps can't tell us about the role of pension and they may just reflect who receives the. Comparing work-pensioners to non-recipients biases pensions to look better than they are while the effect is opposite when comparing social-pension recipients to non-recipients. When boothe are pooled together into a single "any pension", these two distortions push in opposite directions and partly cancel. Since the social pensioners outnumber the work pensioners, the pooled figures end up closer to theirs. Because of these reasons, Section 4.3 estimates them separately.

4.2 Pension receipt and depression

Table 2: Pension receipt and depression

	(1)	(2)	(3)	(4)	(5)
Receives any pension	0.60*	0.18	-0.10	-0.08	0.55
	(0.32)	(0.33)	(0.34)	(0.34)	(0.35)
Age, sex, household structure	No	Yes	Yes	Yes	Yes
Urban, schooling, caste, religion	No	No	Yes	Yes	Yes
MPCE quintiles	No	No	No	Yes	Yes
State fixed effects	No	No	No	No	Yes
Mean of outcome	7.60	7.60	7.60	7.60	7.60
Observations	29,627	29,627	29,627	29,627	29,627
R-squared	0.000	0.004	0.012	0.013	0.032

Standard errors in parentheses

Linear probability models. The outcome is depressed (CIDI-SF) x 100,

so coefficients are in percentage points. Standard errors clustered by household.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 2 shows linear probability models. The outcome is multiplied by 100, which is why the coefficients are in percentage points. Each column adds a group of controls. Column 1 does not account for any controls. Pension recipients are 0.60 percentage points more likely to be depressed, and this is significant only at the 10 percent level. Column 2 adds controls for age, sex and household structure and the estimate falls to 0.18 but it isn't significant. Column 3 adds urban residence, schooling, caste and religion, and the estimate remains insignificant. Columns 4 and 5 add MPCE quintiles and state fixed effects respectively. From columns 3-5, the estimate varies between -0.10 and 0.55, however it remains insignificant throughout. The standard errors are between 0.32 and 0.35 in every column.

My preferred estimate is column 5 because it has all the controls as well as state fixed effects. The estimate is 0.55 percentage points (SE 0.35, p=0.11) and it is not significant. The 95% confidence interval is $[-0.13, 1.24]$. I will now compare the confidence interval with the mean of 7.60 percent given in Table 2. The lower bound of -0.13 is approximately 2 percent of the mean,

so recipients are at most 0.13 percentage points less likely to be depressed, and any meaningful protective association is ruled out. The upper bound of 1.24 is about 16 percent of the mean, so a possibly modest positive association cannot be ruled out. Section 4.3 splits work-related and social pensions.

4.3 Work-related and social pensions

Table 3: Work-related and social pensions

	(1) By type	(2) By scheme	(3) BPL control	(4) BPL households
Work-related pension	-0.27	-0.25	-0.02	-0.11
	(0.58)	(0.58)	(0.59)	(1.24)
Social pension	0.70*		0.57	0.40
	(0.39)		(0.40)	(0.53)
National old-age pension		0.82**		
		(0.42)		
Widow pension		-0.16		
		(0.78)		
Other social pension		0.60		
		(1.51)		
BPL or Antyodaya card			1.57***	
			(0.40)	
Mean of outcome	7.60	7.60	7.70	8.17
Observations	29,627	29,627	28,948	15,243
R-squared	0.032	0.032	0.032	0.042

Standard errors in parentheses

All columns include the controls and state fixed effects of Table 2, column 5.

Outcome x 100. Standard errors clustered by household.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 3 splits pensions by type, and Figure 1 plots the estimates. Each column of Table 3 has the full controls as well as state fixed effects. Column one splits pension into work pension and social pension. The estimate for work pension is −0.27 (SE 0.58) and it is not significant. For social pension, the estimate is 0.70 (SE 0.39) and it is significant at the 10 percent level. This tells us that social-pension recipients are more likely to be depressed than similar non-recipients. The difference between the work-pension and social-pension estimates (−0.97, SE 0.67, p=0.15) is not statistically significant, which means that the data cannot tell the two apart.

Column 2 further splits social pension into the National Old Age Pension Scheme, the widow pension and other social pension schemes. Table 3 shows that only the National Old Age Pension estimate of 0.82 (SE 0.42) is significant. It is significant at the 5 percent level. The estimates

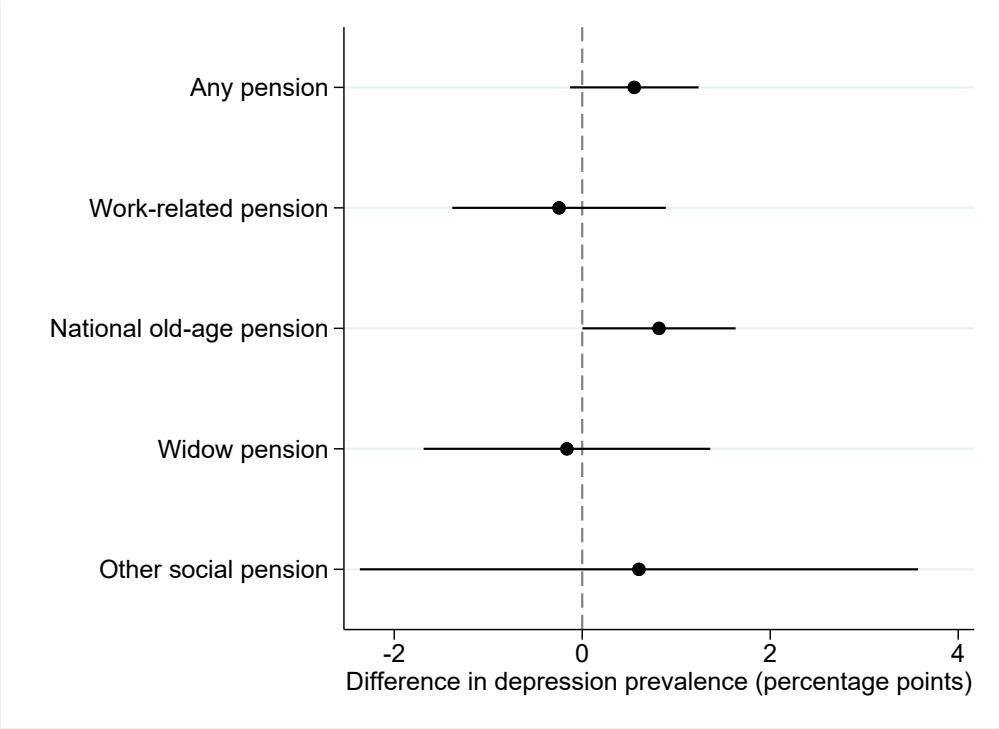


Figure 1: Pension coefficients from the preferred specification. "Any pension" is from Table 2, column 5; the remaining estimates are from Table 3, column 2. Bars are 95% confidence intervals.

for widow pension, other pensions and work pension are −0.16 (SE 0.78), 0.60 (SE 1.51), −0.25 respectively. None of them are significant. So the positive social-pension association comes from the old-age pension, the scheme that requires a BPL household. Figure 1 shows that its interval is the only one that doesn't cross zero.

Column 3 adds an indicator for a BPL or Antyodaya card to the specification in column 1. The estimate for BPL card is 1.57 (SE 0.40) and it is significant at the 1 percent level. BPL households are 1.57 percentage points more likely to be depressed. So, the eligibility condition for the old-age pension is itself a strong predictor of depression. With the BPL card controlled, the social estimate of 0.57 is not significant. In column 4, I look at work-related and social pensions within BPL households. Neither pension estimate is significant. The sample size for column 3 is 28,948 because ration-card information is missing for some respondents, and it is 15,243 in column 4. The significant positive association is attached to the only scheme that requires BPL status, and BPL status itself strongly predicts depression. This is consistent with recipients being selected on an eligibility condition that also predicts depression, so it is not evidence that pensions raise depression.

4.4 Robustness and heterogeneity

Appendix Table 4 checks whether the preferred estimate depends on the choices made in Section 3. Clustering standard errors by village or urban ward instead of by household leaves the estimate at

0.55 and raises the standard error only from 0.35 to 0.36. The broader definition of pension receipt, which adds respondents who expect a work-related pension in future, gives 0.51 (SE 0.34), and a probit model gives an average marginal effect of 0.46 (SE 0.34), so the linear probability model is not driving the result. Dropping MPCE and household structure, which pension receipt may affect, raises the estimate to 0.85 (SE 0.35), significant at the 5 percent level. The largest change comes from using LASI's national weights: the estimate rises to 1.68 (SE 0.62), significant at the 1 percent level. As Solon et al. (2015) point out, a large gap between weighted and unweighted estimates suggests that the association differs across the groups that LASI samples at different rates. None of these checks overturns the main conclusion. The two significant estimates are both positive, and across all six specifications the lower bound of the 95 percent confidence interval is never below about -0.2, so no specification allows a protective association larger than about 0.2 percentage points.

Appendix Table 5 asks whether the association differs across groups, by interacting pension receipt with indicators for women, urban residence, age 70 and above, and holding a BPL card. None of the interaction terms is significant: they are -0.51 (SE 0.63) for women, 0.27 (SE 0.66) for urban residence, 0.02 (SE 0.65) for respondents aged 70 and above, and 0.24 (SE 0.66) for BPL households. There is therefore no reason to believe that the association differs along these dimensions, although with standard errors of about 0.65 these tests can detect only large differences. For the same reason, the analysis cannot identify where the gap between the weighted and unweighted estimates comes from.

5 Interpreting the estimates

These estimates are conditional associations, not causal effects. Pension receipt is not randomly assigned: people qualify through their employment histories or through programme eligibility rules, so recipients may differ from non-recipients in ways the controls do not capture.

The likely direction of the bias differs between the two kinds of pension. Work-related pensions go to people with long careers in formal employment. Such careers bring higher lifetime income, financial literacy and social networks, which MPCE and schooling capture only imperfectly and which plausibly protect mental health. This positive selection pushes the estimate toward lower depression among recipients. Social pensions work the other way. The old-age pension requires a below-poverty-line household and the widow pension requires widowhood, and both conditions are associated with depression: in Table 3, holding a BPL card alone is associated with a 1.57 percentage point higher probability of being depressed. If the controls do not fully capture these disadvantages, this negative selection pushes the estimate toward higher depression among recipients. Because social pensions account for about five in six recipients, this bias is likely to weigh more heavily in the pooled estimate.

Two further sources of bias apply to both kinds of pension. The first is health. The regressions do not control for physical health, because health may itself respond to pension income. But poor

health can lead both to earlier retirement, and hence to receiving a work-related pension rather than still expecting one, and to depression. This would push the estimate toward higher depression among recipients. The second is measurement. The CIDI-SF is a screening instrument rather than a clinical diagnosis, and stigma and imperfect recall may lead respondents to underreport symptoms. If underreporting differs across groups, for example by schooling, it could bias the estimate in either direction. Pension receipt may also be misreported, and random misreporting of a binary variable biases its coefficient toward zero.

A causal answer requires variation in pension receipt that is unrelated to recipients' circumstances. Three sources look promising. The first is the eligibility rules themselves: the national old-age pension is available to BPL households from age 60, and because LASI interviews adults from age 45, BPL respondents just above and just below 60 could be compared, with non-BPL respondents as a check, in the spirit of Galiani et al. (2016). The second is variation across states in pension amounts, which Kaushal (2014) uses to study household welfare. The third is later waves of LASI, which will show how depression changes when the same individuals begin to receive a pension.

6 Conclusion

This paper asked whether older adults in India who receive a pension are less likely to be depressed. Using LASI Wave 1, I find no evidence that they are. Conditional on demographic and socioeconomic controls and state fixed effects, the estimated difference in depression between pension recipients and non-recipients is small and insignificant (0.55 percentage points; 95 percent confidence interval -0.13 to 1.24), ruling out any meaningful protective association. This small pooled estimate combines two very different groups. Work-related pensions go to the educated and better off, while social pensions go, by design, to poor households and widows. Among the separate estimates, the only significant ones are positive and attach to social pensions, especially the national old-age pension, whose eligibility condition, a BPL ration card, is itself strongly associated with depression.

These results do not show that pensions fail to improve mental health. They show that cross-sectional comparisons of recipients and non-recipients cannot settle the question, because who receives a pension is shaped by employment histories and eligibility rules that are themselves linked to mental health. The most promising route to a causal answer is to use those rules directly, by comparing BPL households on either side of the old-age pension's eligibility age, and to follow the same individuals across later waves of LASI as they begin to receive a pension.

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A Additional tables

Table 4: Robustness of the pension coefficient

	(1) Preferred	(2) Weighted	(3) Village SE	(4) Broad	(5) Lean	(6) Probit
Receives any pension	0.55	1.68***	0.55	0.51	0.85**	0.46
	(0.35)	(0.62)	(0.36)	(0.34)	(0.35)	(0.34)
Observations	29,627	29,627	29,627	29,627	29,627	29,627

Standard errors in parentheses

(1) Table 2, column 5. (2) LASI national weights. (3) Clustered by village/ward.

(4) Adds expected work-related pensions. (5) Drops MPCE and household structure.

(6) Probit average marginal effect. Coefficients in percentage points.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 5: Heterogeneity in the pension coefficient

	(1) Women	(2) Urban	(3) Aged 70+	(4) BPL card
Receives any pension	0.81*	0.46	0.54	0.37
	(0.45)	(0.44)	(0.44)	(0.49)
Pension x group	-0.51	0.27	0.02	0.24
	(0.63)	(0.66)	(0.65)	(0.66)
Observations	29,627	29,627	29,627	28,948

Standard errors in parentheses

Group in column heading interacted with pension receipt; full controls as in Table 2, column 5.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

B Variable construction

Depression (CIDI-SF). Depressed-mood pathway (screen MH201, symptom items asked if the episode lasted most of the day, nearly every day, MH202–MH203): (1) loss of interest in most things (MH204); (2) fatigue or low energy (MH205); (3) loss or gain of appetite (MH206–MH207, counted once); (4) trouble concentrating (MH208); (5) feeling worthless (MH209); (6) thoughts of death (MH210); (7) trouble sleeping (MH211). Loss-of-interest pathway (screen MH214, criteria MH215–MH216): (1) fatigue or low energy (MH217); (2) loss of appetite (MH218); (3) trouble concentrating (MH219); (4) feeling worthless (MH220); (5) thoughts of death (MH221); (6) trouble sleeping (MH222).

Pension receipt. Work-related: WE412, "Please tell me, whether you are currently receiving pension or expect to receive pension in future," asked if the respondent currently works in or has retired from the organized sector (WE004 = 1 or WE402 = 1); 1 = currently receiving, 2 = expected in future, 3 = neither. Social: SW202, asked for each scheme the respondent is aware of (SW201 = 1): (a) National Old Age Pension Scheme; (b) widow pension (women only); (c) Annapurna (excluded, as it provides food grains); (d) and (e) other pension schemes.

Ration card. CO301 (household has a ration card) and CO302 (card type: BPL, Antyodaya, APL, or other), from the household questionnaire. The indicator equals one for BPL or Antyodaya cards and zero for APL, other, or no card.